EXNO: 5	Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides
DATE:	

Test Case 1 :

You are experimenting with a rectangular waveguide to analyze the relationship between its dimensions and wave propagation. The waveguide dimensions are a=4.0 cm, b=2.0 cmand the operating frequency is f=8 GHz.

1. Calculate the cutoff frequencies for TE_{10}modes.

2. Predict the phase and group velocities for the TE_{10}mode at the given frequency. 3. If the width of the waveguide is reduced to a=3.0 cm, how does it affect the cutoff frequency and propagation characteristics?

AIM: To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a=4.0 cm, b=2.0 cmand the operating frequency is f=8 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

CALCULATION

PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= "); n=input("enter the value of n= ");
                                    m= "); n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
```

```
                                        9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^
  sp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

Case 1:

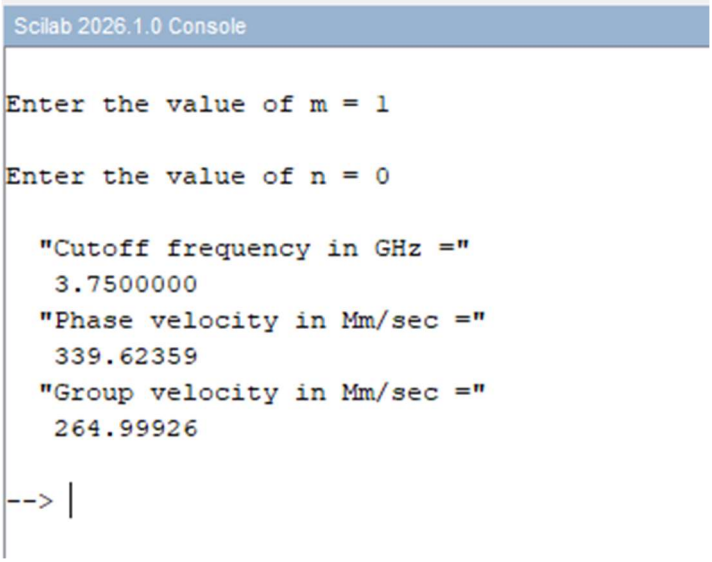
Original Dimensions (a=4.0 cm,b=2.0 cm)
cm,b=2.0 cm)

Inputs:

- m=1, n=0.
- f=8 GHz.

Outputs:

Parameter	Simulation Results	Theoretical Results
	Value	Value
Cutoff Frequency (fc)	3.7500000	
Phase Velocity (vp)	339.62359	
Group Velocity (vg)	264.99926	



Case 2:

Reduced Width (a=3.0 cm,b=2.0 cm)
Inputs:

- m=1, n=0.
- f=8 GHz.

Outputs:

Parameter	Simulation Results	Theoretical Results
	Value	Value
Cutoff Frequency (fc)	5.	
Phase Velocity (vp)	384.30757	
Group Velocity (vg)	234.18742	

```
Scilab 2026.1.0 Console
--> exec('D:\TRANSMISSION\EXP 3 TC1.sci', -1)
Enter the value of m = 1

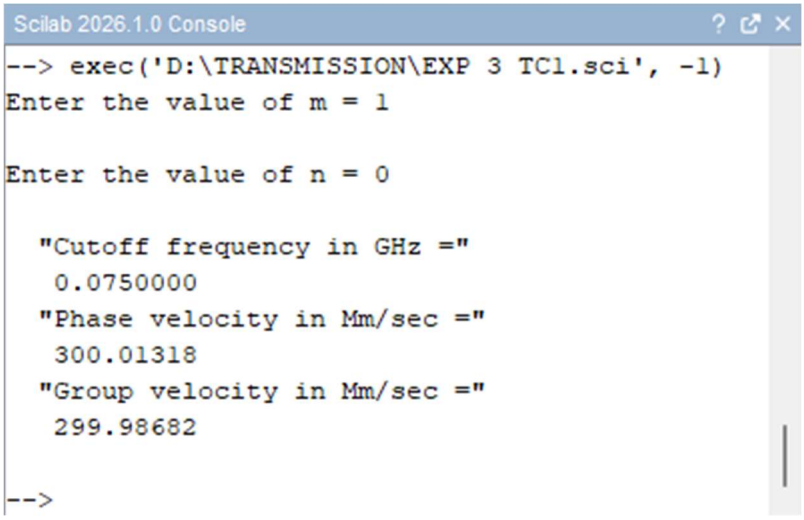
Enter the value of n = 0

  "Cutoff frequency in GHz ="
   5.
  "Phase velocity in Mm/sec ="
   384.30757
  "Group velocity in Mm/sec ="
   234.18742

-->
```

Table: Variation of Phase and Group Velocities with a

Waveguide Width (a)	Simulation Results			Theoretical Results		
	Cutoff Frequency (fc)	Phase Velocity (v_p)	Group Velocity (v_g)	Cutoff Frequency (fc)	Phase Velocity (v_p)	Group Velocity (v_g)
4.0 cm	3.7500000	339.62359	264.99926			
3.5 cm	0.046875	300.00515	299.99485			
3.0 cm	0.0500000	300.00586	299.99414			
2.5 cm	0.06	300.00844	299.99156			
2.0 cm	0.0750000	300.01318	299.98682			



RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

Test Case 2 :

In a radar system operating at =15 GHz the waveguide dimensions are = 2.5 and = 1.25 cm 1. Compute the cutoff frequencies for TE_{10} modes.

2. Determine the phase and group velocities for the TE_{10} mode.

3. If dispersion is critical to radar signal accuracy, what strategies would you recommend maintaining signal fidelity?

AIM: To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a= 2.5 cm, b=1.25 cmand the operating frequency is f=15 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

CALCULATION

PROGRAM:

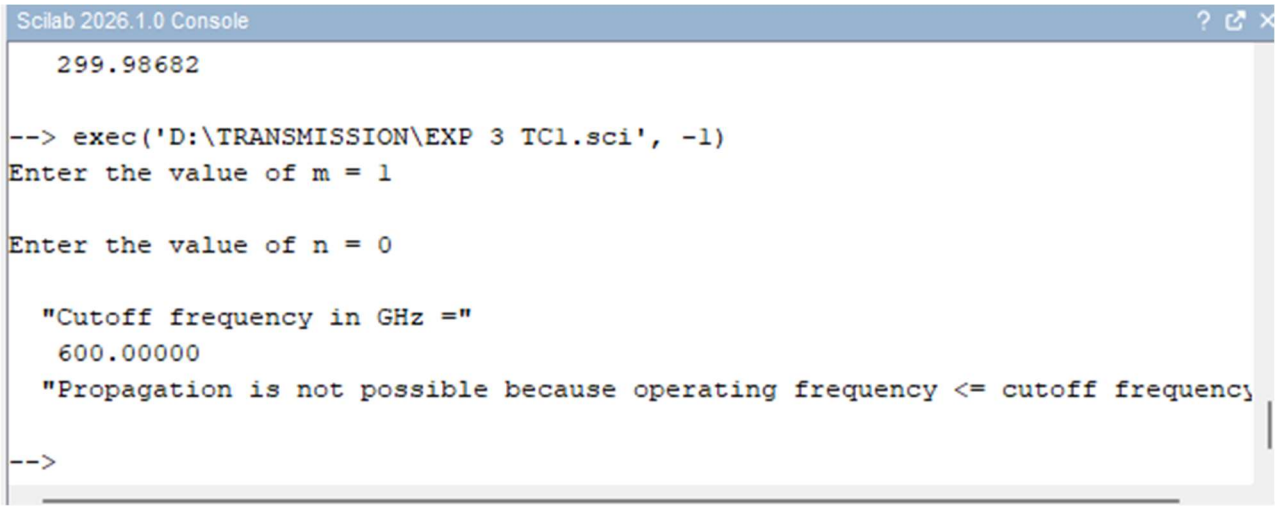
```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
```

```
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= "); n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

Program Output for TE_{10} Mode:

Using the program with a=0.025 m b=0.0125 m and f=15 GHz:

Parameter	Simulation Results	Theoretical Results
	Value	Value
Cutoff Frequency (f_c)	600.00000	
Phase Velocity (v_p)		
Group Velocity (v_g)		



RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

Test Case 3 :

A waveguide with dimensions a=5.0 cm and b=2.5 cmis used to carry a signal at f=10 GHz

> **1. Determine the number of TE modes that can propagate at this frequency. 2. For the TE_{10}mode, calculate the phase velocity and group velocity. 3. Discuss how the presence of multiple propagating modes affects the transmission quality and suggest practical solutions to limit multimode effects.**

AIM:

To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a= 5.0 cm, b=2.5 cmand the operating frequency is f=10 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

CALCULATION

PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
```

```
disp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

Waveguide Parameters:

- Width (a) = 5.0 cm=0.05m
- Height (b) = 2.5 cm=0.025 m
- Operating Frequency (f) = 10 GHz
- Speed of Light (V) = 3×10^8 m/s

1. ***Number of TE Modes that Can Propagate:***

The cutoff frequency for a TE_{mn}_mode is:

$f_c(m,n)=V/2$ (sqrt $[(m/a)^2+(n/b)^2]$)

Modes propagate if f>fc(m,n).The number of modes can be determined by calculating fc for different combinations of m and n.

	Simulation Results		**Theoretical Results**	
Mode (m,n)	**Cutoff Frequency (fc)**	**Propagation (Yes/No)**	**Cutoff Frequency (fc)**	**Propagation (Yes/No)**
TE_{10}	6.	YES		
TE_{01}	6.	YES		
TE_{11}	6.7082039	YES		
TE_{20}	6.	YES		
TE_{21}	8.4852814	YES		
TE_{02}	12.	NO		

```
Scilab 2026.1.0 Console

Enter the value of m = 0

Enter the value of n = 1

  "Cutoff frequency in GHz ="
   6.
  "Phase velocity in Mm/sec ="
   375.
  "Group velocity in Mm/sec ="
   240.

. -->
```

```
Scilab 2026.1.0 Console

Enter the value of m = 0

Enter the value of n = 2

  "Cutoff frequency in GHz ="
   12.
  "Propagation is not possible because f <= fc."

--> s
```

Result:

Five TE modes can propagate TE_{10}, TE_{01}, TE_{11}, TE_{20}, TE_{21}, TE_{02}

2. Output for TE_{10} Mode:

Using the program with a=5 m b=2.5 m and f=10 GHz:

Parameter	Simulation Results	Theoretical Results
	Value	Value
Cutoff Frequency (f_c)	0.0300000	
Phase Velocity (v_p)	300.00135	
Group Velocity (v_g)	299.99865	

RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

Test Case 4 :

A research lab is experimenting with high-order TE modes (m,n>1) in a waveguide with dimensions a=3.0 cm b=1.5 cm, and an operating frequency of f=12 GHz. 1. Calculate the cutoff frequencies for TE_{10} mode.

2. **Compute the phase and group velocities of this mode.**
3. **Discuss how high-order modes can be utilized for advanced applications such as mode multiplexing in communications.**

AIM:

To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a=3.0 cm, b=1.5 cmand the operating frequency is f=12 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

Waveguide Parameters:

· Width (a) = 3.0 cm=0.03 m
· Height (b) = 1.5 cm=0.015 m
· Operating Frequency (f) = 12 GHz=12×10^9 Hz
· Speed of Light (V) = 3×10^8 m/s

1. No. of Modes:

To analyze all modes, calculate fc for other modes with m,n>1.

OUTPUT:

	Simulation Results		Theoretical Results	
Mode (m,n)	**Cutoff Frequency (fc)**	**Propagation (Yes/No)**	**Cutoff Frequency (fc)**	**Propagation (Yes/No)**
TE_{10}	5.			
TE_{20}	10.			
TE_{01}	10.			
TE_{11}	11.180340			
TE_{21}	14.142136	NO		
TE_{30}	15.000000	NO		

```
Enter the value of m = 2

Enter the value of n = 0

  "Cutoff frequency in GHz ="
   10.
  "Phase velocity in Mm/sec ="
   542.72042
  "Group velocity in Mm/sec ="
   165.83124
```

```
  Enter the value of m = 2

  Enter the value of n = 1

    "Cutoff frequency in GHz ="
     14.142136
    "Propagation is not possible because f <= fc."

  -->
```

2. Program Output for TE_{10} Mode:

Using the program with a=0.03 m b=0.015 m and f=12 GHz:

Parameter	Simulation Results	Theoretical Results
	Value	**Value**
Cutoff Frequency (f_c)	5.	
Phase Velocity (v_p)	330.01146	
Group Velocity (v_g)	272.71780	

RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.